



Unemployment Rate Forecasting: A Hybrid Approach

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Abstract

Unemployment has always been a very focused issue causing a nation as a whole to lose its economic and financial contribution. Unemployment rate prediction of a country is a crucial factor for the country's economic and financial growth planning and a challenging job for policymakers. Traditional stochastic time series models, as well as modern nonlinear time series techniques, were employed for unemployment rate forecasting previously. These macroeconomic data sets are mostly nonstationary and nonlinear in nature. Thus, it is atypical to assume that an individual time series forecasting model can generate a white noise error. This paper proposes an integrated approach based on linear and nonlinear models that can predict the unemployment rates more accurately. The proposed hybrid model of the unemployment rate can improve their forecasts by reflecting the unemployment rate's asymmetry. The model's applications are shown using seven unemployment rate data sets from various countries, namely, Canada, Germany, Japan, Netherlands, New Zealand, Sweden, and Switzerland. The results of computational tests are very promising in comparison with other conventional methods. The results for asymptotic stationarity of the proposed hybrid approach using Markov chains and nonlinear time series analysis techniques are given in this paper which guarantees that the proposed model cannot show 'explosive' behavior or growing variance over time.

Keywords Unemployment rate · ARIMA model · Autoregressive neural networks · Hybrid model · Asymptotic stationarity

1 Introduction

Economic indicators such as GDP and labor statistics are used by investors to forecast economic trends and decide on the appropriate investment policies. In particular, the unemployment rate for any country represents one of the most important

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economic indicators for financial market participants due to its correlation with the country's business cycle and its influence on the monetary policy (Blanchard and Leigh 2013). Accurate forecasting of the unemployment rate is central to economic decision-making and the design of policy-making to recognize early the socio-economic problems so that reduction of the same can be planned. The study of unemployment rates and macroeconomic forecasting started blooming in the middle of the 1990s. Many time series models have been employed extensively for the prediction of macroeconomic variables, including unemployment. Previous studies on unemployment rate suggested an asymmetry in the unemployment rate data for various European countries (Milas and Rothman 2008). One of the primary time-series implications of such behavior is that it is inconsistent with a linear data generating process with symmetrically distributed innovations.

In previous studies on unemployment rate forecasting for various developed countries, the autoregressive integrated moving average (ARIMA) model was applied for analyzing Germany and Spain's unemployment data obtained from online search (Funke 1992; Vicente et al. 2015). The usefulness and effectiveness of the classical linear ARIMA model were evident from the results obtained while using various European unemployment rate forecasting data sets (Edlund and Karlsson 1993; Dumičić et al. 2015) and out-of-sample forecasts for Canadian unemployment rates (Khan Jaffur et al. 2017). But the situation was a bit different in case of unemployment rate forecasting for the USA. The threshold autoregressive (TAR) model, a classical nonlinear time series model, outperformed the linear time series models for forecasting the USA unemployment rate data set (Montgomery et al. 1998). For short term forecasting of seasonally adjusted monthly USA unemployment data sets, nonlinear models outperform the linear models (Proietti 2003; Nagao et al. 2019). The current progress in the area of modern statistics and machine learning have equipped the forecasters with nonlinear forecasting tools such as artificial neural networks (ANN), deep learning, and support vector machines (SVM) among many others (Katris 2019; Atsalakis et al. 2007). ANN is found to be the most accurate in forecasting unemployment over the asymmetric business cycle for the USA, Canada, UK, France, and Japan (Moshiri and Brown 2004; Peláez 2006; Wang and Zheng 2009). The previous results show that the nonlinear models are well-versed to seize the asymmetry of unemployment rate time series for long-term forecast horizons (Feuerriegel and Gordon 2019). Still, there remains an asymmetry in unemployment rate forecasting and its elimination is bound to be challenging (Galbraith and van Norden 2019).

The classical ARIMA model is competitive for forecasting stochastic time series whereas nonlinear ANN has produced favorable results in the past decades. Nevertheless, neural nets have the apparent drawback, viz., finding the 'optimal' network architecture. To overcome this drawback, the autoregressive neural network (ARNN) model was proposed in the recent literature (Faraway and Chatfield 1998). ARNN is a "white-box-like" model that fits a feed-forward neural net having one hidden layer to any time series data set with lagged values of the series as inputs (Teräsvirta et al. 2005). It has the advantages of less complexity and easy interpretability over ANN formulation (Hyndman and Athanasopoulos 2018). The data sets at hand contain both linear and nonlinear patterns in the

current problem of unemployment rate forecasting. It will be critical for policy-makers to make any decision based on a single model since one can see regular changes in the dynamic behavior of the unemployment rates. By hybridizing linear and nonlinear models, one can reduce the bias and variances of the prediction error of component models (Oliveira and Torgo 2014). Thus, combining both the linear and nonlinear models will be preferred for accurately model such complex autocorrelation structures (Khashei and Bijari 2011; Firmino et al. 2014). Several hybrid models were applied in the past to solve various forecasting problems that arose in the stock market, financial econometrics, electricity, epidemiology, and other applied areas (Zhang 2003; Pai and Lin 2005; Aladag et al. 2009; Chakraborty et al. 2019; Chakraborty and Ghosh 2020). All these hybrid models are practically shown to be useful in solving real-life forecasting problems, but the theoretical result for asymptotic stationarity is not proved for any of these hybrid models - up to our knowledge.

This paper proposes a hybrid approach that studies the relationship between linear and nonlinear components of the unemployment rate time series. The proposed hybrid methodology assumes an additive relationship between linear and nonlinear models, assuming that different models can capture the linear and nonlinear patterns of a time series separately, and then the forecasts can be combined. The proposed hybrid method will be appropriate for explaining variations of the unemployment rate in the presence of nonstationarity and nonlinearity in this time series. In the first phase of the proposed model, an ARIMA model is applied to catch the linear patterns of the data set. Residual error values of the ARIMA model are calculated and restored for further modeling. In the next stage, a nonlinear ARNN model is applied to capture the nonlinear trends in the data set using the residual values obtained from ARIMA. We call this two-step approach as ‘hybrid ARIMA–ARNN’ model. Of particular interest for the statisticians in time series forecasting literature is the question of asymptotic stationarity of the model, viz. whether the probabilistic structure of the series is constant over time or at least asymptotically constant. In this study, we have shown the asymptotic stationarity of the proposed hybrid ARIMA–ARNN model using Markov chains and nonlinear time series analysis techniques. The theoretical results for asymptotic stationarity guarantee that the model cannot have a growing variance with time. The asymptotic behavior of the proposed hybrid model is especially crucial for predictions over a larger interval of time or when using the model to generate an artificial time series. The theoretically proven asymptotic stationarity for the hybrid approach will fill the gap between theory and practice when hybrid time series models are considered. Finally, we test the performance of the proposed model on seven unemployment rate data sets and compared our proposed model with several other state-of-the-art forecasting models.

The rest of the paper is organized as follows. In Sect. 2, we describe the unemployment rate data sets and the development of the methodology. The asymptotic stationarity of the proposed model is discussed in Sect. 3. The experimental results are presented in Sect. 4. Finally, the economic implications of this study are given in Sect. 5.

2 Material and Methods

2.1 Unemployment Data and Its Characteristics

The unemployment rate represents the number of unemployed as a percentage of the labor force. Forecasting unemployment rate can be defined as the projected value for the number of unemployed people as a percentage of the labor force. Six seasonally adjusted monthly data sets on unemployment rates for Canada, Germany, Japan, Netherlands, Sweden, and Switzerland and one quarterly data of New Zealand were collected from the open-access data repository FRED Economic Data sets (Link: <https://fred.stlouisfed.org/>) and OECD data repository (Link: <https://data.oecd.org/>).

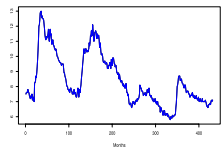
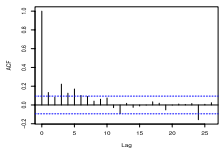
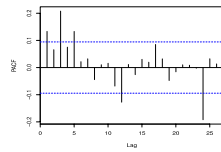
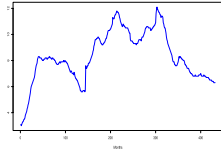
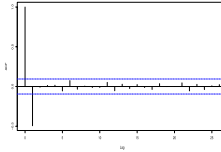
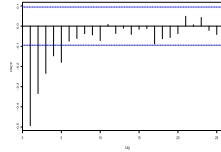
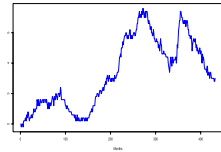
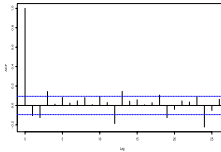
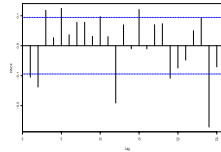
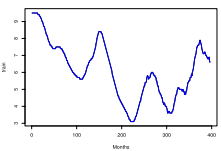
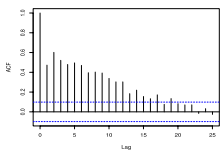
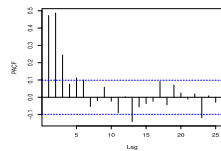
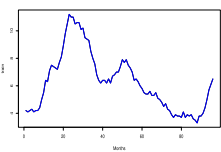
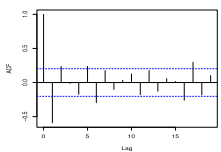
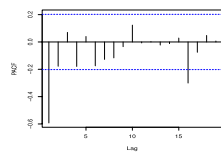
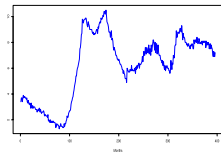
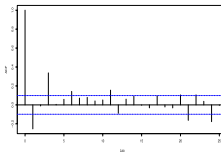
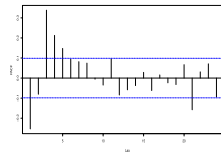
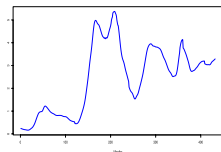
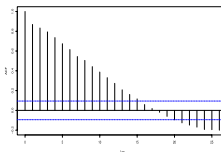
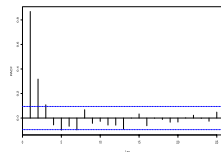
A summary of these unemployment rate data sets is available in Table 1. The plots of the training data for various countries are given in Table 2.

The graphical presentation of the data sets confirms the presence of nonstationarity and nonlinearity in these unemployment rate data. We also use a well-known linearity test, namely the new-F test, since it covers the most extensive set of alternatives of nonlinearity (Terui and Van Dijk 2002). The new-F test rejects the null hypothesis of linearity most strongly for all the seven unemployment rate data sets. In this work, we consider approaches for the prediction of univariate unemployment rate time series. A combination of linear and nonlinear methods that take into account the specific characteristics of data can offer more accurate predictions. Some inherent characteristics of these data sets are the departure from normality and the nonlinearity of the dependence structure of the data which are also evident from previous studies of these data sets (Galbraith and van Norden 2019; Katris 2019; Nagao et al. 2019). To take into account the nonstationarity, the linear ARIMA model is first considered. Furthermore, since nonlinearity exists, the neural networks-based ARNN model seems suitable in the second stage of the hybrid model.

Table 1 Descriptions of the unemployment rate data sets

Data	Total size	Maximum value	Minimum value	Training set size (years)	Test set size (years)
Canada	468	13.0	5.6	432 (1980–2015)	36 (2016–2018)
Germany	468	12.1	3.0	432 (1980–2015)	36 (2016–2018)
Japan	468	5.80	1.9	432 (1980–2015)	36 (2016–2018)
Netherlands	432	9.5	3.1	396 (1983–2015)	36 (2016–2018)
New Zealand	132	11.2	3.3	100 (1986–2009)	32 (2010–2018)
Sweden	432	10.5	1.3	396 (1983–2015)	36 (2016–2018)
Switzerland	468	5.37	0.1	432 (1980–2015)	36 (2016–2018)

Table 2 Graphical analysis of training unemployment rate data sets for different countries and its corresponding ACF and PACF plots

Country	Training data	ACF plot	PACF plot
Canada			
Germany			
Japan			
Netherlands			
New Zealand			
Sweden			
Switzerland			

2.2 Forecast Evaluation Metrics

The performance of different forecasting models are evaluated based on mean absolute error (MAE), root mean square error (RMSE), and mean absolute percent error

(MAPE) for the unemployment rate data sets (Ahmed et al. 2010). The mathematical expressions of these performance evaluation metrics are as follows.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}; \quad MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|; \quad MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|,$$

where y_i is the actual output, $\hat{y}_i$ is the predicted output, and n denotes the number of data points. By definition, the lower the value of these performance metrics, the better is the performance of the concerned forecasting model.

2.3 Methodology

This work proposes a hybrid model based on ARIMA and ARNN model to forecast the unemployment rates for seven countries. Below we first discuss about the constituent models to be used in the hybridization.

2.3.1 Autoregressive Integrated Moving Average (ARIMA) Model

ARIMA is a linear time series model, used for tracking linear tendencies in stationary time series data. ARIMA model is denoted by $ARIMA(p, d, q)$. The parameters p and q are the order of the AR model and the MA model respectively, and d is the level of differencing (used for converting nonstationary series into a stationarity one). ARIMA model can be mathematically expressed as follows:

$$y_t = \theta_0 + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j},$$

where y_t denotes the actual value of the variable under consideration at time t , ε_t is the random error at time t , ϕ_i and θ_j are the coefficients of the model. The necessary steps for building an ARIMA model for any given time series data set are as follows: model identification of the model (achieving stationarity), estimation of model parameters (the autocorrelation function (ACF) and the partial autocorrelation function (PACF) plots are used to select the AR and MA model parameters, respectively), and model diagnostics checking [finding the ‘best’ fitted forecasting model using Akaike information criterion (AIC) or the Bayesian information criterion (BIC)] (Hyndman and Athanasopoulos 2018).

2.3.2 Autoregressive Neural Network (ARNN) Model

ANN is a widely used supervised learning model, highly useful for sophisticated nonlinear time series forecasting. Any neural net architecture can be described as a network of “neurons”, arranged in layers, namely the input layer, hidden layer, and output layer. The information from one layer to another layer is passed using weights that are selected using a risk minimization based ‘learning algorithm’. The ARNN model is a modified neural network model especially designed for time series data

sets which uses a pre-specified number of hidden neurons in its architecture (Faraway and Chatfield 1998). It uses lagged values of the time series as inputs to the model. ARNN(p, k) is a nonlinear feed-forward neural net model with one hidden layer (having p lagged inputs) and k hidden units in the hidden layer. It also uses BIC as the criterion for comparing different models created by ARNN. Here $\hat{x}_t$ is computed using selected past observations $x_{t-j_1}, \dots, x_{t-j_p}$ as the inputs. Thus, ARNN model with one hidden layer can be depicted with the following mathematical form:

$$\hat{x}_t = \phi_0 \left\{ w_{c_0} + \sum_k w_{k_0} \phi_k \left(w_{c_k} + \sum_i w_{ik} x_{t-j_i} \right) \right\},$$

where $\{w_{c_k}\}$ denotes the connecting weights and ϕ_i is the activation function. Weights of the ARNN model are trained using a gradient descent backpropagation algorithm (Rumelhart et al. 1985). The ARNN(p, k) model uses p as the number of lags for an AR(p) model and k is usually set to $k = \left\lceil \frac{(p+1)}{2} \right\rceil$ for non-seasonal time series data (Hyndman and Athanasopoulos 2018).

2.3.3 Proposed Hybrid ARIMA–ARNN Model

We propose a hybrid ARIMA–ARNN model, which is a two-step pipeline approach. In the first step of the proposed hybrid approach, an ARIMA model is built to model the linear components of time series, and a series of forecasts are generated. In the second phase, the ARIMA residuals are modeled using a nonlinear ARNN model. The formulation of the proposed hybrid ARIMA–ARNN model (Z_t) can be formally represented as follows:

$$Z_t = Y_t + N_t,$$

where Y_t is the linear part and N_t is the nonlinear part of the hybrid model. We can estimate both Y_t and N_t from the training data set. Let, $\hat{Y}_t$ be the forecast value of the ARIMA model at time t and ε_t represent the error residuals at time t , obtained from the ARIMA model. We can then write

$$\varepsilon_t = Z_t - \hat{Y}_t.$$

The residuals are modeled by the ARNN model and can be represented as follows

$$\varepsilon_t = f(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-n}) + \zeta_t, \quad \text{for some integer } n,$$

where f is a nonlinear function of the ARNN model and ζ_t is the random shocks. Therefore, we can write the combined forecast as:

$$\hat{Z}_t = \hat{Y}_t + \hat{N}_t,$$

where $\hat{N}_t$ is the forecasted value of the ARNN model. ARNN models the left-over autocorrelations in the residuals, which ARIMA could not model. This is important because due to the model mis-specification and disturbances in the unemployment

rate time series, the linear ARIMA model may fail to generate white noise behavior in the forecast residuals. Therefore, if the error series is modeled again, the performance of the original forecaster can be improved, even though marginally at times. The algorithmic representation of the proposed hybrid approach is also given in Algorithm 1.

Algorithm 1: Proposed hybrid ARIMA-ARNN approach

- 1 Given a time series, input the in-sample (training) and out-of-sample (test) data.
 - 2 Determine the best ARIMA(p, d, q) model using the in-sample (training) data.
 - ARIMA parameters p , d , and q values are selected using standard procedures, as described in Section 2.3.1.
 - Obtain the predictions using the selected ARIMA(p, d, q) model for the training data.
 - Obtain the residual series (ε_t) by subtracting ARIMA predicted values from the original training series.
 - 3 Determine the best ARNN(p, k) model on the training residual series (ε_t).
 - Perform lag selection for the training of residual series and apply ARNN model with p selected lagged inputs from (ε_t) and k hidden units as described in Section 2.3.2.
 - Obtain predictions using the ARNN model ($\hat{\varepsilon}_t$).
 - 4 Final predictions ($\hat{Z}_t$) are obtained by combining the ARIMA predictions with ARNN predictions ($\hat{\varepsilon}_t$).
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3 Asymptotic Stationarity of the Proposed Model

ARIMA has an in-built mechanism to transform a nonstationary time series into a stationary time series by taking the differencing of the given time series (Brockwell and Lindner 2010). ANNs are asymptotically stationary and it requires the process to be stationary while training the neural network, but when applied to a nonstationary process, the out of sample predictions becomes poor (Leoni 2009). In the hybrid formulation based on ARIMA and ARNN model, this problem can be overcome since we deal with only the additive error terms generated by ARIMA and model it using ARNN model.

Let us consider a nonlinear ARNN model generated by additive noise of the ARIMA model. Let ε_t denote a time series generated by a nonlinear autoregressive process as defined in (1). Thus, the stochastic difference equation is of the form:

$$\varepsilon_t = f(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-p}, \theta) + \zeta_t, \quad (1)$$

where ζ_t is an i.i.d. noise process and $f(\cdot, \theta)$ is a feedforward neural network with weight parameter vector θ . This is called an ARNN process of order p and has k hidden nodes in its one hidden layer, denoted by ARNN (p, k) model. ARNN models are a natural generalization of the classic linear autoregressive AR(p) process

$$\varepsilon_t = \alpha_1 \varepsilon_{t-1} + \dots + \alpha_p \varepsilon_{t-p} + \zeta_t.$$

Now, we define f as a neural network and z denote a p -dimensional input feature to the ARNN model (error residuals obtained from ARIMA model). We consider the following architecture:

$$f(z) = c_0 + \sum_{i=1}^k w_i \sigma(a_i + a_i' z) \quad (2)$$

where a_i , w_i , and c_0 are scalar weights, a_i are p -dimensional weight vectors and $\sigma(\cdot)$ is a bounded nonlinear sigmoid function such as tan-hyperbolic or logistic function. Since we consider unbounded additive noise terms generated by the ARIMA model, we can not apply the results based on state space model of the Markov chains. But if we suppose $E(\zeta_t) = 0$, then f equals the conditional expectation $E(\varepsilon_t | \varepsilon_{t-1}, \dots, \varepsilon_{t-p})$ and $f(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-p})$ is the best prediction for ε_t in the mean square sense. Thus, for the unbounded noise terms, we define the following notation:

$$\begin{aligned} z_{t-1} &= (\varepsilon_{t-1}, \dots, \varepsilon_{t-p})' \\ F(z_{t-1}) &= (f(z_{t-1}), \varepsilon_{t-1}, \dots, \varepsilon_{t-p+1})' \\ e_t &= (\zeta_t, 0, \dots, 0)' \end{aligned}$$

Then we write scalar AR(p) models in (1) as a first-order vector model

$$z_t = F(z_{t-1}) + e_t \quad (3)$$

with $z_t, e_t \in \mathbb{R}^p$ (Chan and Tong 1985). Also, we write

$$\begin{aligned} p^n(z, A) &= P\{z_{t+n} \in A | z_t = z\} \\ p(z, A) &= p^1(z, A) \end{aligned}$$

for the probability of going from point z to set $A \in \mathbb{B}$ in n steps, then $\{z_t\}$ with $p(z, A)$ forms a Markov chain with state space $(\mathbb{R}^p, \mathbb{B}, \cdot)$, where $\mathbb{B}$ is a Borel set on $\mathbb{R}^p$ and $\cdot$ be the usual Lebesgue measure.

The Markov chain $\{z_t\}$ is said to be φ -irreducible, if for some σ -finite measure φ on $(\mathbb{R}^p, \mathbb{B}, \cdot)$ and for all $z \in \mathbb{R}^p$, we have

$$\sum_{n=1}^{\infty} p^n(z, A) > 0$$

whenever $\varphi(A) > 0$. This essentially means that all parts of the state space can be reached by the Markov chain irrespective of the starting point. Another interesting property of the Markov chain is *aperiodicity* that loosely means that there are no (infinitely often repeated) cycles (Tong 1990).

Since we are interested in the long-term properties of the time series, one may ask whether certain features such as mean or variance change over time or remain constant. A time series is called weakly stationary if $E(\varepsilon_t) = \mu$ and $cov(\varepsilon_t, \varepsilon_{t+h}) = \gamma_h \forall t$, viz., mean and covariance do not depend on time t . A more strong criterion is that

the whole distribution of the process does not depend on the time and then the series is called strictly stationary. It is interesting to note that strong stationarity implies weak stationarity if the second order moments of the series exist (Brockwell and Lindner 2010). But we further need to define asymptotic stationarity in this context:

Definition 1 If ε_t is strictly stationary then $P(\varepsilon_t \in A) = \Pi(A)$, for all t and $\Pi(\cdot)$ is called the stationary distribution of the series. We call the series asymptotically stationary if it converges to its stationary distribution (if it is not started with Π):

$$\lim_{t \rightarrow \infty} P(\varepsilon_t \in A) = \Pi(A).$$

Also, trivially the series can only be stationary from the beginning if it starts with the stationary distribution such that $\varepsilon_0 \sim \Pi$.

Definition 2 Let $\{z_t\}$, a Markov chain, is said to be geometrically ergodic if there exists a probability measure $\Pi(A)$ on the state space $(\mathbb{R}^p, \mathbb{B}_p)$, and for $\rho > 1$ and for all $z \in \mathbb{R}^p$,

$$\lim_{n \rightarrow \infty} \rho^n \|p^n(z, \cdot) - \Pi(\cdot)\| = 0$$

where $\|\cdot\|$ denotes the total variation. Then Π satisfies the invariance equation

$$\Pi(A) = \int p(z, A) \Pi(dz), \quad \forall A \in \mathbb{B}_p.$$

If the markov chain is geometrically ergodic then its distribution will converge to Π and the corresponding time series is called asymptotically stationary (Trapletti et al. 2000).

Lemma 1 Let $\{z_t\}$ is defined by (3), and let $E|\zeta_t| < \infty$ and the probability distribution function (PDF) of ζ_t is positive everywhere in $\mathbb{R}$. Then if f is defined by (2), the Markov chain $\{z_t\}$ is φ -irreducible and aperiodic.

Proof It can easily be shown that $\{z_t\}$ is φ -irreducible if the support of the PDF of ζ_t is the whole real line, viz., the PDF is positive everywhere in $\mathbb{R}$ (Chan and Tong 1985). In our case, every non-null p -dimensional hypercube can be reached in p steps with positive probability (and hence every non-null Borel set A).

A necessary and sufficient condition for $\{z_t\}$ to be *aperiodic* is to have a set A and positive integer n such that $p^n(z, A) > 0$ and $p^{n+1}(z, A) > 0$ for all $z \in A$ (Tong 1990, p. 455). In this case, this is true for all n due to consideration of the unbounded additive noise. $\square$

Remark 1 Lemma 1 states that the state space of the Markov chain cannot be reduced depending on the starting point. An example of a reducible Markov chain is a series that is always positive if only $z_0 > 0$ (and negative otherwise). But this cannot happen in the ARNN(p, k) model due to having unbounded additive noise terms.

The theorem below states the necessary condition for geometric ergodicity of a markov chain is due to (Chan and Tong 1985).

Theorem 1 (Chan and Tong 1985) *Suppose $\{\varepsilon_t\}$ and $\{z_t\}$ are defined as in (1) and (3), respectively. Further, let F be compact (or preserve compact set) and can be decomposed as $F = F_h + F_d$, and the following conditions hold:*

- (i) $F_h(\cdot)$ is continuous and homogeneous and $F_d(\cdot)$ is of bounded range;
- (ii) The origin is a fixed point of F_h and F_h is uniform asymptotically stable.
- (iii) If $E|\zeta_t| < \infty$ and PDF of ζ_t is positive everywhere in $\mathbb{R}$; then $\{z_t\}$ is geometrically ergodic.

Proof See (Chan and Tong 1985, pp. 671–673). □

The next theorem gives the main result for asymptotic stationary of the hybrid ARIMA–ARNN model.

Theorem 2 *Let $E|\zeta_t| < \infty$ and the PDF of ζ_t is positive everywhere in $\mathbb{R}$, and $\{\varepsilon_t\}$ and $\{z_t\}$ are defined as in (1) and (3), respectively. Then if f is a nonlinear neural network as defined in (2), then $\{z_t\}$ is geometrically ergodic and $\{\varepsilon_t\}$ is asymptotically stationary.*

Proof The noise process ζ_t satisfies $E|\zeta_t| < \infty$ by assumption (e.g., Gaussian noise). It is also important to note that neural network activation functions, more precisely logistics or tan-hyperbolic activation functions, are continuous compact functions and have bounded range. For application to the unemployment rate problem, we considered the logistic activation function $F(z) = \frac{1}{1+e^{-z}}$ in the ARNN model while implementation.

Thus $\{z_t\}$ satisfies all the criteria to be geometrically ergodic and using Theorem (1), one can write that for the ARNN process with $F_h \equiv 0$ and $F_d \equiv F$. Thus, the series $\{\varepsilon_t\}$ is asymptotically stationary. □

Remark 2 We have found sufficient conditions for asymptotic stationarity of the proposed hybrid ARIMA–ARNN model. This is important for predictions over larger intervals of time, for example, one might train the network on an available sample and then use the trained network to generate new data with similar properties like the training sample. Theoretical results on asymptotic stationary guarantees that the proposed approach cannot have ‘explosive’ behavior (or growing variance with time).

4 Results and Discussion

Seven unemployment rate data sets are divided into training and testing data, as described in Table 1. These data sets are mostly nonlinear and non-Gaussian in nature and statistical tests confirm this (see Sect. 2.1). Even time series plots of the data sets show nonlinearity and non-stationarity (refer to Table 2). We experimentally evaluate the performances of ARIMA, ANN, SVM, ARNN model, hybrid ARIMA–ANN model (Zhang 2003), hybrid ARIMA–SVM model (Pai and Lin 2005), in comparison with our proposed hybrid ARIMA–ARNN model for all these data sets.

We start the experimental evaluation with the classical ARIMA(p, d, q) using the “forecast” package in R statistical software. Fitting an ARIMA model, we need to specify the orders of the model. Using the ACF plot and PACF plots, we can decide the value of the model’s parameters. We performed the ADF test for stationarity check for the model to determine the amount of d in the ARIMA model. The ‘best’ fitted ARIMA model is chosen using AIC value for each training data set. As the ARIMA model is fitted, predictions are generated for one year and three year time steps. We also compute training data predicted values and calculate the residual errors. In the second stage, residuals obtained using the ARIMA model are remodeled with ARNN(p, k) model. We employ a pre-defined Box-Cox transformation set to $\lambda = 0$ to ensure the forecast values to stay positive. The values of p and k are obtained by training the network, which is a data-dependent approach as in (Hyndman and Athanasopoulos 2018). Further, both the linear and nonlinear forecasts are added together to get the final forecasts. Other individual models like SVM were implemented using the “e1071” package in R statistical software. ARNN was applied using the “nnfor” package with “nnetar” function and for ANN model, we have used “caret” package with the “mlp” function in R. For all the experiments with the ANN model, we have used one hidden layer with the number of hidden neurons $k \approx \sqrt{n}$, where n is the training sample size.

The experimental results are obtained as follows: ARIMA (3,1,3) was fitted to Canada unemployment rate data with AIC and log-likelihood values as -154.25 and 84.37 , respectively. The ARIMA residuals were further trained with ARNN (5,3) model with an average of 20 networks. Further, we computed the predicted test outputs of the hybrid ARIMA–ARNN model and compared it with actual test outputs. The values of different performance metrics are reported in Table 3. For the Germany data set, ARIMA (1,2,1) having AIC = -494.05 and $L = 250.03$ was fitted. An ARNN (1,1) model (1–1–1 network configuration) with an average of 20 networks, each having four weights, was trained on ARIMA training residuals. Both the forecasted results of ARIMA and ARNN are added together to obtain the final forecast values based on which RMSE, MAE and MAPE values are computed and report them in Table 4. ARIMA (2,1,4) was fitted to Japan data having AIC = -688.11 and log-likelihood (L) value as 351.05 . Further, the model’s residuals were trained using ARNN (25,13) model with an average of 20 networks, each of which is a 25–13–1 network with 352 weights. Then the forecast results of ARIMA along with ARNN residual forecasts are added together

to obtain the final forecast values. Finally, we compute RMSE, MAE and MAPE and report them in Table 5.

Similarly, we applied the proposed hybrid ARIMA–ANN model for New Zealand and Netherlands data sets and the results are reported in Tables 7 and 6, respectively. ARIMA (1,1,2) with log-likelihood = -22.54 and $AIC = 53.07$ was first fitted to the quarterly unemployment rate data set of New Zealand. The residuals are modeled with an ARNN (10,5) model with an average of 20 networks. For the Netherlands' monthly data, we fit an ARIMA (1,2,1) and ARNN (12,6) on ARIMA residuals. In the case of Sweden data, ARIMA (3,1,1) was fitted with $AIC = 0.65$ and $L = 4.68$. Further, ARIMA residuals were trained using ARNN (16,8) model (16–8–1 network configuration) with an average of 20 networks, each having 145 weights. Again, we obtain the final predicted forecast values for Sweden's test data sets by adding both the ARIMA and ARNN forecasts. The values of the performance metrics for measuring forecasting accuracy are reported in Table 8. Finally, ARIMA (2,1,2) was fitted to Switzerland data having $AIC = -1745.46$ and L equals to 877.73. The residuals obtained from the ARIMA model was trained using ARNN (3,2) model with an average of 20 networks, each of which is a 3–2–1 network with 11 weights. Finally, the forecast results of ARIMA and ARNN residual forecasts are added together to obtain the predicted forecasted values. We then compute the values of the performance measures, viz. RMSE, MAE and MAPE values and report them in Table 9.

For comparison purposes, we applied single ARIMA, ANN, ARNN, SVM, along with hybrid ARIMA–ANN (Zhang 2003), hybrid ARIMA–SVM model (Pai and Lin 2005) models for the seasonally adjusted unemployment rate data sets of these countries. All the experimental results are reported in Tables 3, 4, 5, 6, 7, 8, and 9. The predicted forecasts for the test data sets of the proposed hybrid model for seven data sets, along with actual test values, are plotted in Fig. 1. The performances of the proposed hybrid ARIMA–ARNN model are superior as compared to all the individual models. In comparison to other hybrid models, other than the Japan unemployment rate data set, our proposal outperformed all the hybrid models in the significant margin. The theoretically proven asymptotic stationarity of the proposed hybrid model suggested that the model can not have a growing variance over time. The consistency in experimental results empirically approves the same. Thus, the usefulness of the proposed methodology is experimentally validated as well.

5 Economic Implications and Conclusions

Forecasting the unemployment rate is very important for financial market participants and is a reliable indicator of labor market conditions. The release of the monthly (quarterly) unemployment rate for a country is one of the most important regular economic events for market participants. The effect of the unemployment news on stock returns is not so straightforward because of the relative importance of information on labor market conditions and monetary policy changes over time depending on the state of the economy. Considering the difficulty in predicting market reactions, forecasting the unemployment rate

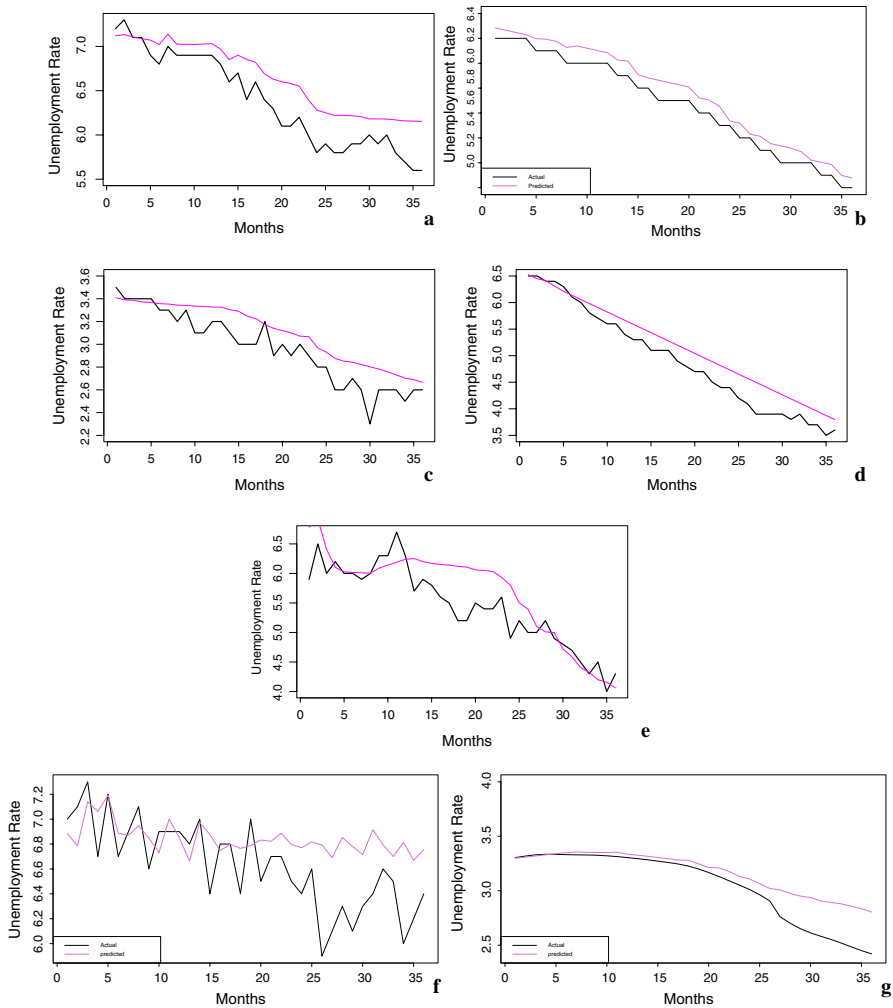


Fig. 1 Actual versus predicted (based on hybrid ARIMA–ARNN model) forecasts for the test data sets of the Canada (a), Germany (b), Japan (c), Netherlands (d), New Zealand (e), Sweden (f) and Switzerland (g) unemployment rate data sets

accurately is highly useful for investors to hedge the market risk arising from the unexpected change in business conditions and monetary policy.

In this study, we proposed a hybrid ARIMA–ARNN model using an error modeling approach that performs considerably well for unemployment rates forecasting for the countries like Canada, Germany, Japan, Netherlands, New Zealand, Sweden and Switzerland. The proposed hybrid ARIMA–ARNN model filters out linearity using the ARIMA model and predicts nonlinearities present in the error residuals with an ARNN model. The proposed hybrid ARIMA–ARNN model can explain the linear and nonlinear tendencies present

Table 3 Performance metrics for different forecasting models on the Canadian unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.133	0.115	1.623	0.847	0.685	9.708
ANN	0.137	0.117	1.095	0.837	0.614	9.365
ARNN	0.126	0.113	1.084	0.801	0.613	9.247
SVM	0.273	0.248	1.915	0.998	0.740	10.92
Hybrid ARIMA–SVM	0.145	0.135	1.135	0.835	0.711	9.595
Hybrid ARIMA–ANN	0.118	0.108	1.017	0.638	0.615	8.387
Proposed hybrid ARIMA–ARNN	0.106	0.098	0.838	0.627	0.601	8.017

Bold values suggest that they are best results

Table 4 Performance metrics for different forecasting models on the Germany unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.098	0.092	1.490	0.361	0.303	5.177
ANN	0.127	0.120	4.295	0.564	0.505	7.394
ARNN	0.104	0.099	6.783	0.569	0.533	6.365
SVM	0.101	0.099	1.594	0.566	0.509	6.272
Hybrid ARIMA–SVM	0.090	0.089	1.537	0.360	0.305	5.120
Hybrid ARIMA–ANN	0.082	0.096	1.558	0.306	0.297	4.243
Proposed hybrid ARIMA–ARNN	0.077	0.071	1.068	0.300	0.291	4.156

Bold values suggest that they are best results

Table 5 Performance metrics for different forecasting models on the Japan unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.300	0.272	6.985	0.586	0.502	14.621
ANN	0.260	0.229	6.518	0.469	0.413	11.627
ARNN	0.269	0.236	6.713	0.483	0.426	11.934
SVM	0.221	0.192	5.902	0.469	0.421	10.334
Hybrid ARIMA–SVM	0.226	0.187	5.377	0.386	0.336	9.722
Hybrid ARIMA–ANN	0.191	0.172	4.987	0.370	0.321	9.317
Proposed hybrid ARIMA–ARNN	0.222	0.186	5.348	0.371	0.326	9.468

Bold values suggest that they are best results

Table 6 Performance metrics for different forecasting models on the Netherlands unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.144	0.119	2.003	0.306	0.270	5.671
ANN	0.228	0.174	2.906	1.304	1.048	17.935
ARNN	0.249	0.192	3.177	0.938	0.784	14.518
SVM	0.228	0.174	2.906	1.304	1.048	17.935
Hybrid ARIMA–SVM	0.145	0.120	2.023	0.308	0.272	5.706
Hybrid ARIMA–ANN	0.143	0.118	2.002	0.306	0.270	5.668
Proposed hybrid ARIMA–ARNN	0.140	0.114	1.192	0.300	0.264	5.529

Bold values suggest that they are best results

Table 7 Performance metrics for different forecasting models on the New Zealand unemployment rate (quarterly) data

Model	3-Year ahead forecast			8-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	1.121	1.118	13.779	1.985	1.850	24.775
ANN	1.630	1.461	18.398	4.796	4.167	40.109
ARNN	1.329	1.280	17.073	1.895	1.797	24.863
SVM	1.329	1.280	17.073	1.895	1.797	23.940
Hybrid ARIMA–SVM	1.059	1.028	14.249	1.978	1.842	25.182
Hybrid ARIMA–ANN	1.019	0.986	13.748	1.941	1.820	24.763
Proposed hybrid ARIMA–ARNN	0.998	0.944	11.272	1.318	1.239	22.992

Bold values suggest that they are best results

Table 8 Performance metrics for different forecasting models on the Sweden unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.194	0.155	2.323	0.384	0.305	4.458
ANN	0.257	0.282	2.895	0.324	0.412	3.797
ARNN	0.253	0.178	3.275	0.324	0.288	4.336
SVM	0.396	0.358	4.253	0.489	0.410	5.525
Hybrid ARIMA–SVM	0.198	0.157	2.252	0.391	0.312	4.544
Hybrid ARIMA–ANN	0.190	0.157	2.244	0.387	0.309	4.502
Proposed hybrid ARIMA–ARNN	0.189	0.151	2.024	0.363	0.298	4.231

Bold values suggest that they are best results

Table 9 Performance metrics for various forecasting models on Switzerland unemployment rate (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.047	0.037	1.095	0.437	0.314	9.365
ANN	0.026	0.023	1.094	0.526	0.433	11.176
ARNN	0.027	0.028	1.715	0.498	0.340	10.924
SVM	0.045	0.035	1.135	0.535	0.511	11.295
Hybrid ARIMA–SVM	0.040	0.038	1.117	0.438	0.315	9.387
Hybrid ARIMA–ANN	0.026	0.024	1.090	0.435	0.310	9.273
Proposed hybrid ARIMA–ARNN	0.026	0.022	1.038	0.427	0.301	8.917

Bold values suggest that they are best results

in the unemployment rate data sets of developed countries better as compared to the traditional single and hybrid models. It also yields better forecast accuracy than various single and hybrid models for most of the data sets considered in this study. The proposal will be useful for economists and policymakers in decision and policy makings. Although any econometric phenomena can fluctuate heavily because of various external factors over time. And those fluctuations are generally challenging to be appropriately captured for accurate forecasting. However, the proposed model may still predict with better accuracy provided the conditions stated in the main result for asymptotic stationarity of the hybrid model are satisfied.

Theory and data agree on a number of analytical results: (a) Over the last four decades, the unemployment rates for most of the countries considered in this study had no consistent trend at all and has asymmetrical cyclical movements; (b) The best short-term forecasts of monthly and quarterly unemployment rate data sets using the proposed hybrid model generated better short-term and long-term forecasts as compared to other competitive methods. An immediate extension of this work is to see the application of the model for seasonal unemployment rate data sets. Another possible extension of the work would be to expand this hybrid approach for multivariate time series forecasting problems that arise in various applied domains.

6 Data and Codes

For the sake of reproducibility of this work, all codes and data sets are made available at <https://github.com/Sayak7873>.

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